Introduction

Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

An illusion stimulus is not the same thing as an illusion report. A stimulus can be physically specified, generated, encoded, and inspected without establishing what every observer will see or hear. This distinction is central to reproducible cognitive science: the physical manipulation must be auditable, while observer-level effects require a task, calibrated presentation conditions, randomization, and data.

Introduction I

Visual illusion classifications are useful maps rather than exhaustive or universally agreed ontologies (Gregory 1997). Auditory and audiovisual families add spectral structure, channel separation, temporal coincidence, and spatial alignment. The Sound-Induced Flash literature illustrates the boundary particularly clearly: the physical contrast is a flash paired with different beep counts, while the perceptual claim requires a listener, timing, and a response task (Hirst et al. 2020; Shams et al. 2000).

DuckRabbit therefore treats an illusion as a typed stimulus-construction problem. Each catalog entry declares modality, mechanism, perceptual signature, cognitive process, requirements, evidence status, output kind, and implementation status. The taxonomy is orthogonal and explicitly provisional; it is an engineering index linked to sources, not a replacement for domain-specific theory.

The release also treats the package as research software rather than as an unannotated collection of media files. Reproducible computational research depends on preserving the code, inputs, environment assumptions, and execution path needed to recreate a result (Sandve et al. 2013; Wilson et al. 2014). The FAIR literature extends that responsibility to algorithms, tools, and workflows, while software-citation guidance emphasizes identifying the exact software object and version used (Wilkinson et al. 2016; Lamprecht et al. 2020;

Smith et al. 2016). DuckRabbit applies these principles narrowly: it makes the construction pipeline and its generated artifacts citable and inspectable, but it does not treat metadata quality as evidence of a perceptual effect.

The package is organized around three falsifiable engineering hypotheses:

Introduction I

1. Identical typed requests produce identical canonical arrays and canonical digests.
2. Encoders and decoders preserve declared media facts within an explicit format-specific tolerance, or verification fails.
3. Parameter manipulations change measurable physical stimulus properties; any observer-level interpretation remains a preregistered hypothesis until data exist.

This boundary has a nineteenth-century scientific lineage. Fechner's *Elemente der Psychophysik* formalized the problem of relating controlled stimulus differences to measured sensation, while Wheatstone's binocular-vision experiments made the distinction between the physical images delivered to two eyes and the resulting depth interpretation experimentally explicit (Fechner 1860; Wheatstone 1838). Helmholtz's physiological optics then integrated measurement of the eye, visual geometry, and theories of perceptual inference (Helmholtz 1867). DuckRabbit does not reproduce those historical experiments; it inherits their methodological lesson that stimulus conditions and observer inferences must be represented as distinct objects.

That lineage has deeper and less geographically narrow roots. Ptolemy's ancient *Optics*, Ibn al-Haytham's medieval Arabic optics, and the Mohist Canon's early Chinese discussions of optics and mechanics show that questions about image formation, geometry, knowledge, and evidence were developed across multiple intellectual traditions (Ptolemy 1996; al-Haytham 1989; Schemmel and Boltz 2022; Dai 2015). Early-modern discussions sharpened the distinction between a physical presentation and an inferred percept: Kircher documented optical display, Berkeley analyzed learned spatial inference, and the Molyneux–Cheselden record made cross-sensory transfer and observer testimony explicit problems (Kircher 1646; Berkeley 1709; Degenaar and Lokhorst 2020; Cheselden 1728). DuckRabbit cites these sources as historical and methodological precedents, not as evidence that its generated files reproduce ancient,

early-modern, or clinical observations.

The present catalog contains 18 entries, of which 17 are implemented, 0 are planned, and 1 require external fixtures. This status vocabulary makes breadth visible without claiming that a finite package covers all known phenomena.